Does chess make children smarter? What the studies found

A large trial in England and the main reviews, in plain words. Checked 8 November 2026.

The short answer

In a trial of 4,009 pupils in England, about half had a year of chess lessons. A year later, the chess pupils did no better in maths, reading or science.

Smaller studies found a boost, but almost none compared chess with another engaging activity, and the few that did found little or no difference. What chess does for certain: it makes children better at chess, and many enjoy it.

Study	Who and how many	What was measured	What it found
Chess in Primary Schools trial (EEF; evaluated by UCL Institute of Education, Jerrim et al., 2016)	4,009 Year 5 pupils in 100 schools in England; about half, in schools chosen at random, had 30 hours of lessons in 2013/14	National tests (Key Stage 2) one year after the lessons ended	No evidence of a positive impact on maths, science or reading (impact: “0 months”). Half the pupils liked the lessons a lot; 8% didn’t. Staff perceived a positive impact on maths
Sala and Gobet (2016), a review	24 studies	Maths, reading and thinking skills	A moderate boost (maths 0.38), but almost none of the studies used an active comparison group
Sala and Gobet (2017), two experiments	233 and 52 primary-school children; 25 hours of chess in the first	Maths problem-solving	Chess against draughts, then against go (the go study is the review’s one active-control exception): differences “not statistically significant” both times
Sala, Foley and Gobet (2017), on the EEF trial	—	The trial’s design	It measured a year later, and many pupils scored near the top of the maths test. “The evidence produced so far is insufficient to establish those benefits either”
Sala and Gobet (2017), “Does Far Transfer Exist?”	Reviews of chess, music and memory training	Whether the skills carry over to other subjects	The better the design, the smaller the effect. The one chess study with an active comparison group found a minimal effect (0.10). “Far transfer of learning rarely occurs”
Sala et al. (2017), a review	Chess players and non-players	Thinking-test scores	Chess players score higher, by “a moderately large effect”. That’s a correlation: it doesn’t show chess caused it

How to read a claim about chess (four questions, from the studies’ own methods)

1. Was there a comparison group of children who didn’t learn chess?
2. Did that group do another engaging activity, or nothing new?
3. Were the children put into the groups at random?
4. When were they tested: straight after the lessons, or later?

The better question

Not “Will it make my child smarter?” but “Will my child enjoy it, and will we play together?”

Sources, with links: educationendowmentfoundation.org.uk (Chess in Primary Schools – trial); doi.org/10.3389/fpsyg.2017.00238; doi.org/10.1177/0963721417712760; doi.org/10.1016/j.edurev.2016.02.002; doi.org/10.3758/s13420-017-0280-3.